

(*) EEE

(23/11/25)

(25/11/25)

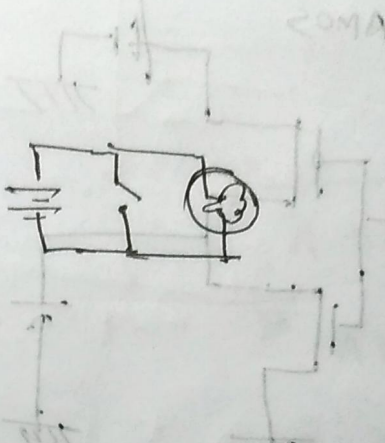
NOR
NAND
next

Gate

① → Transistor

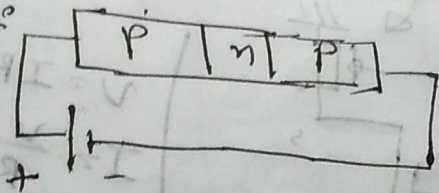


② → VLSI

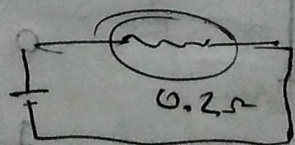
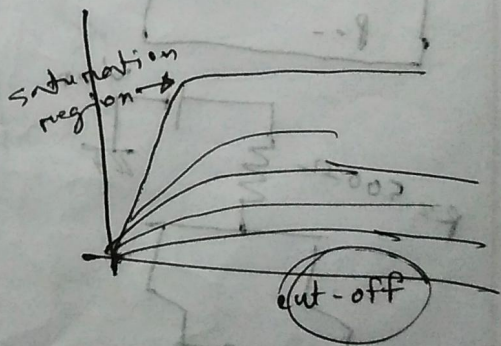
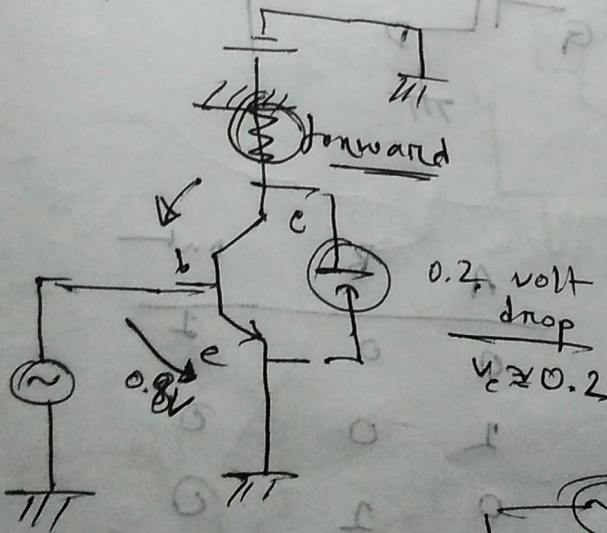


⇒ Audio → quantization → sampling

transistor



- Saturation (2x forward)
- active (forward + reverse)
- reverse → cut off



① ΔV
forward

$$V = IR$$

$$\text{or } I = \frac{V}{R}$$

0 → 1

1 → 0

Inverter

EEE

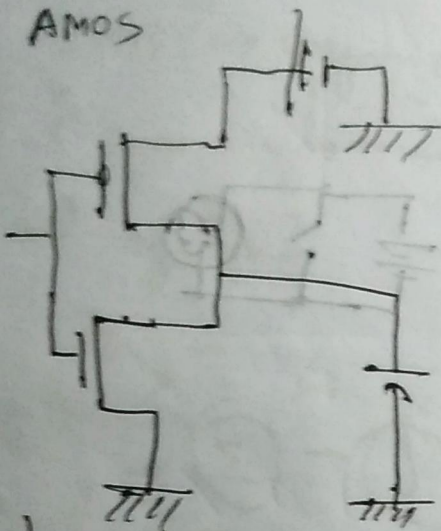
(30/11/25)

(25/11/25)

EEE

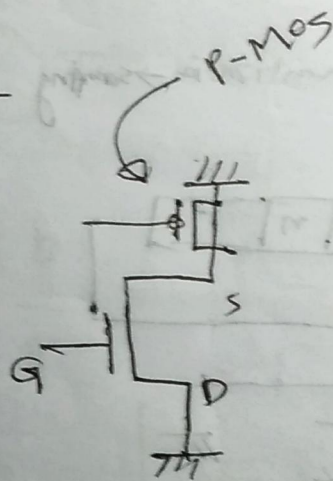
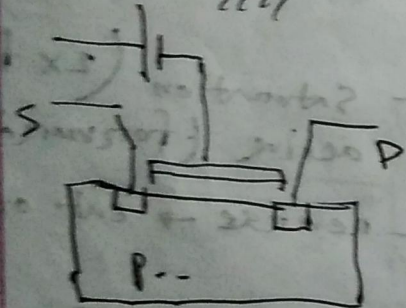
PMOS

AMOS



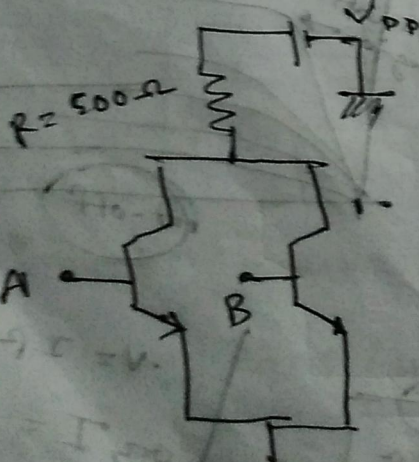
→ Inverter
load resistance
replace

→ MOSFET



$$V = IR$$

$$I = \frac{V}{R}$$



A	B	out
0	0	1
1	0	0
0	1	0
1	1	0

Fan out
1 gate
multiple
proc

V_{LST}

Noise

margin

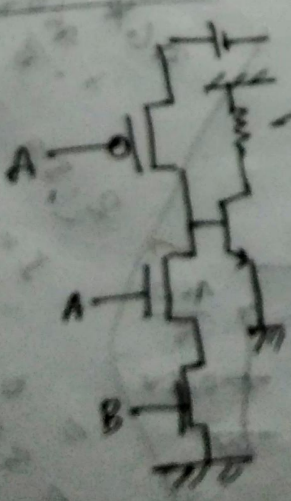
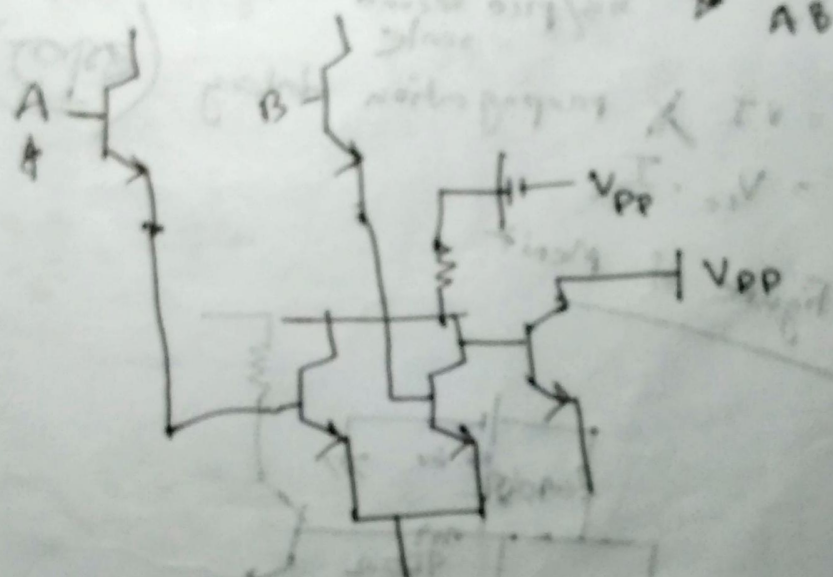
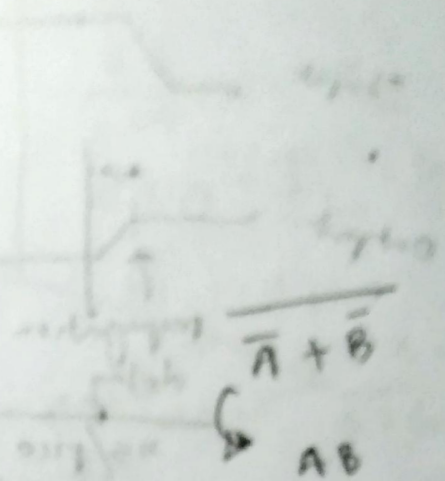
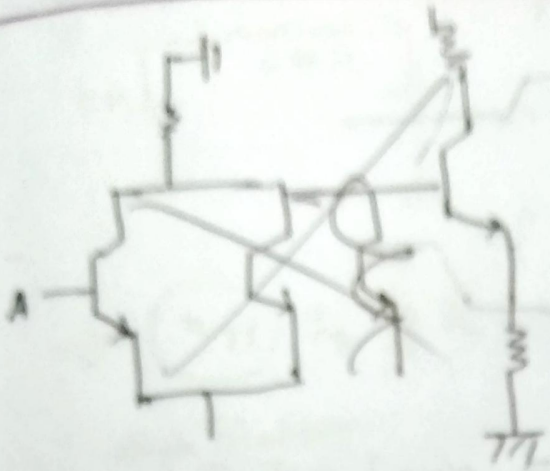
0.5

tolerance

1 level
noise

interference

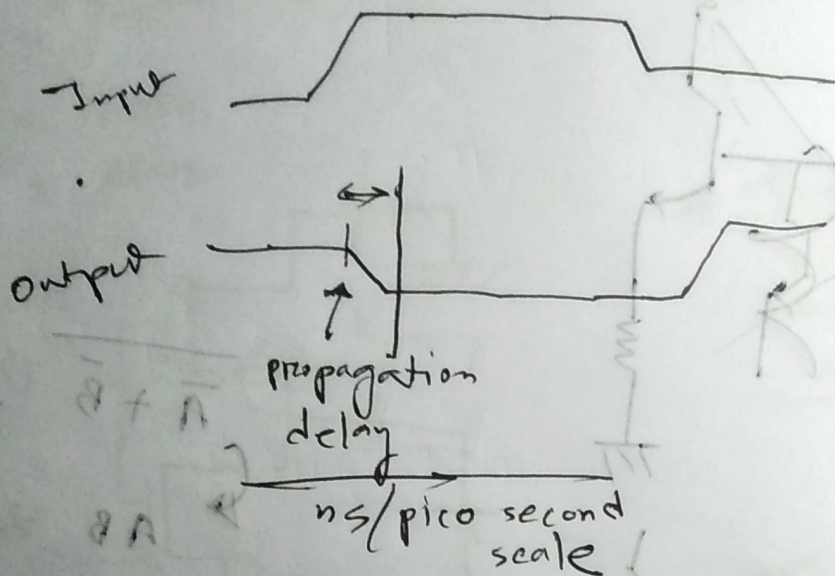
$A \cdot B$ $(A + B)$ $(\overline{A \cdot B})$



For And gate

0	0	1
0	1	1
1	0	1
1	1	0

NAND gate



statics

$$P = VI \times \text{propagation delay}$$

$$= V_{cc} \cdot I_{cc}$$

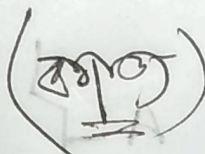
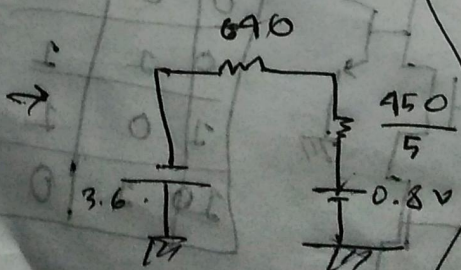
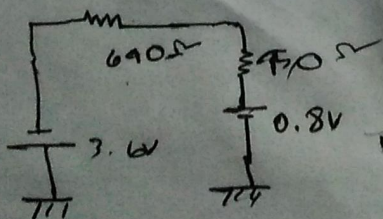
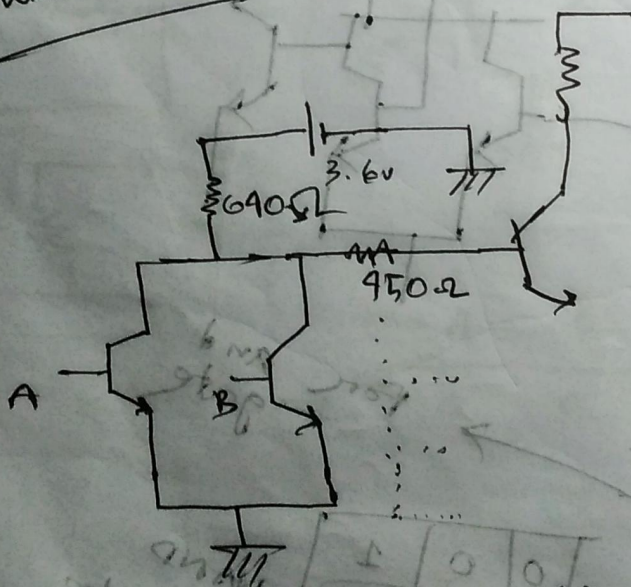


Figure of Merit



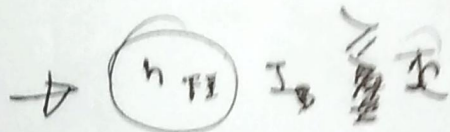
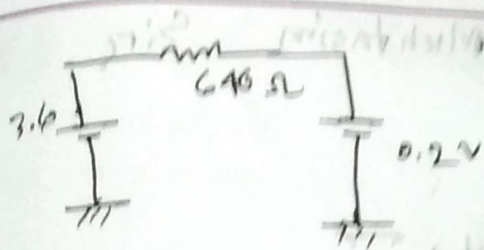
$$3.6V = 640 \times I + 450 \times I + 0.8$$

$$2.8 = 730 I$$

$$I = \frac{2.8}{730}$$

$$I = 0.0038 A$$

$$3.8 mA$$



$$3.6 = 640 I_B + 0.2$$

$$\frac{3.4}{640} = I_B$$

$$5.3 \text{ A} = I_C$$

Amplification factor / gain

$$I_B = 0.76$$

$$I_C = 5.3$$

$$\frac{5.3}{0.76}$$

per single

$$I_B = \frac{3.4}{5} = 0.76$$

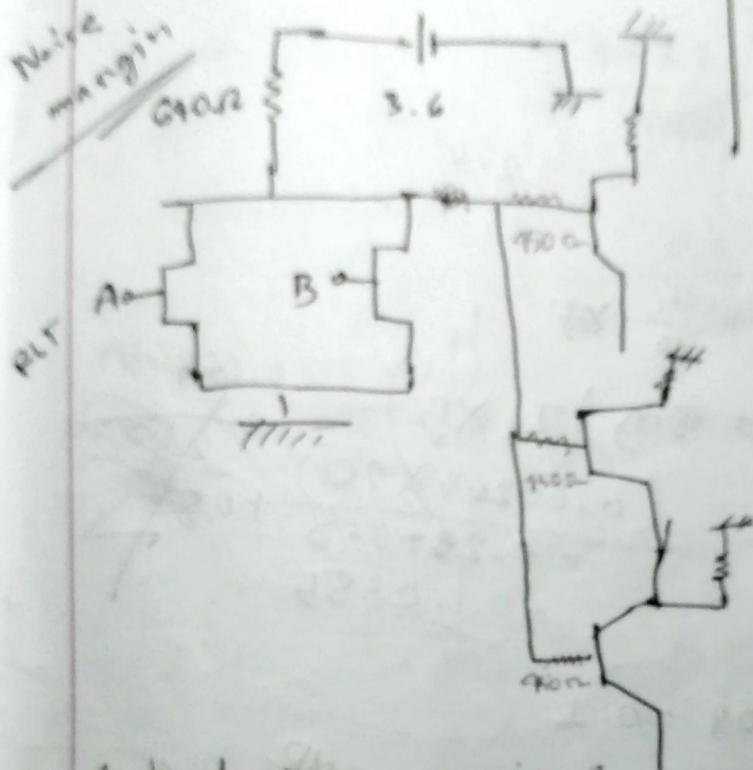
$$I_B = \frac{3.6 - 0.8}{640 - \frac{450}{N}} \cdot \frac{1}{N}$$

$$10 \cdot I_B > 5.3$$

Fan out

→ EEE (Tubid Sin) 3/12/23

Noise margin

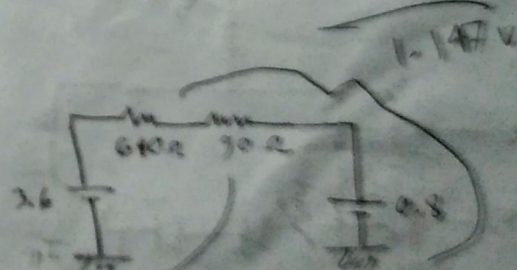
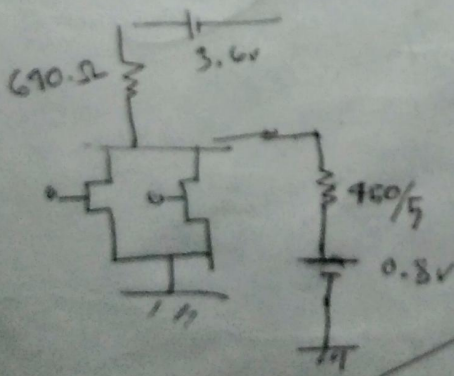


PMOS voltage
gate is 3.6V

noise margin
0.3

$$NM = 0.3$$

1 level noise margin :



$$-3.6 + 690 I_B + 450 I_B + 0.8 = 0$$

$$\text{or, } I_B = 0.0031 \text{ A}$$

$$V = IR = 0.392 + 0.8 = 1.197$$

$$h_{FE} \cdot I_B \approx I_C$$

$$\therefore I_C = \frac{3.6 - 0.2}{640}$$

$$= 0.0053$$

$$h_{FE} = 10$$

$$N = 5$$

$$I_B = \frac{I_C}{10}$$

$$I_{B_{avg}} = (I_B \times 5) = 0.00265A$$

$$\frac{(0.00265 \times 90)}{0.2 + 0.8} + 0.8 = 1.0385$$

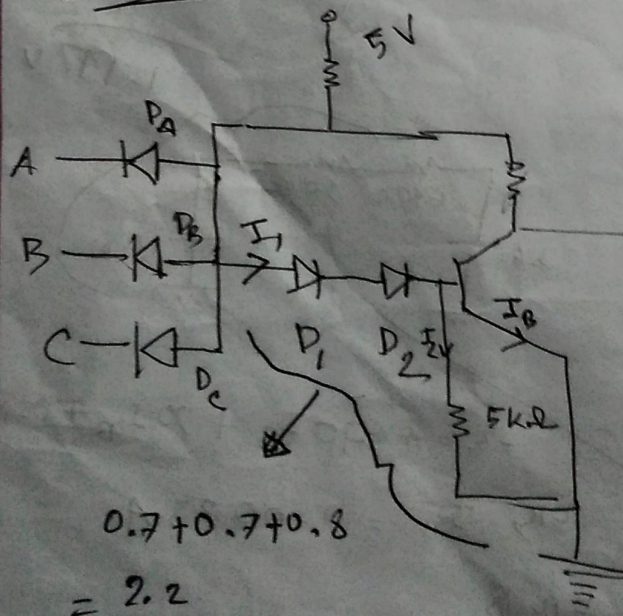
Disadvantage of RTL

$$= \frac{450}{5} A$$

$$0.2V$$

$$\Delta T = 1.19 - 1.09 = 0.1$$

DTL

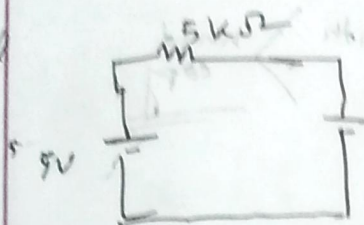


NAND

0	1	1	1
1	1	1	0

EEE - Tonhid Sir (9/11/25)

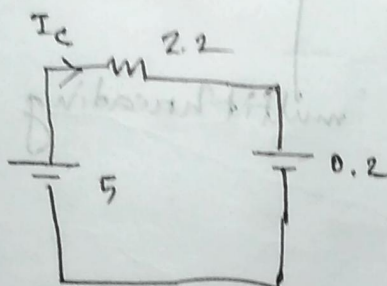
Analysis



$$I_1 = \frac{5 - 2.2}{5 + 2.2} = 0.56$$

$$I_2 = \frac{0.8}{5 + 2.2} = 0.16$$

$$I_B = I_1 - I_2 = 0.56 - 0.16 = 0.4$$



$$I_C = 2.18 \text{ mA}$$

$$h_{FE} \times I_B \gg I_C$$

$$\text{or } h_{FE} \approx 5.45$$

Assume Gain 10

$$I_L = \frac{5 - 0.2 - 0.7}{5} = 0.82 \text{ mA}$$

$$\text{or } 0.82 \times N + 2.18 \leq 10 \times 0.4$$

$$\text{or } 0.82 \times N \leq 4 - 2.18$$

$$\text{or } N = 2.21 \approx 2$$

Noise margin:

$$5 - 2.2 = 2.8$$

$$2.8 + 0.6 = 3.4$$

cutting v. Diode

$$V_P = 0.9$$

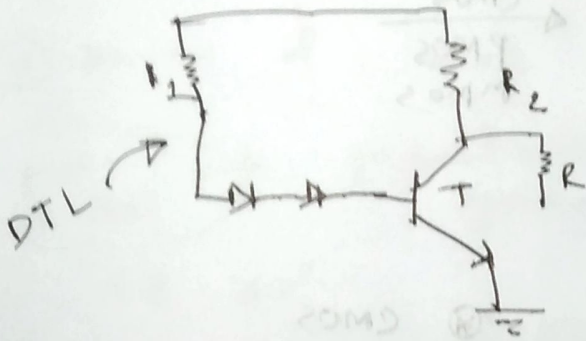
$$0.6 + 0.6 + 0.5 = 1.7$$

$$\therefore \text{1 level margin} = 0.8$$

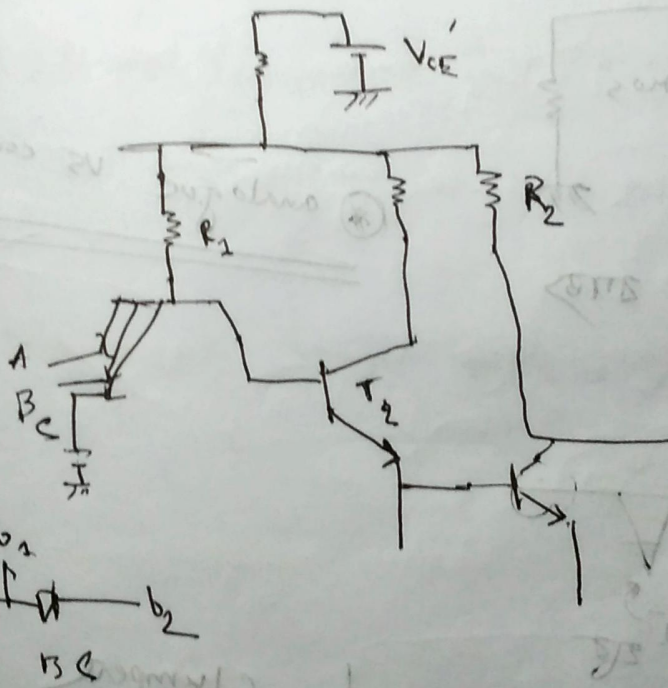
level

gap

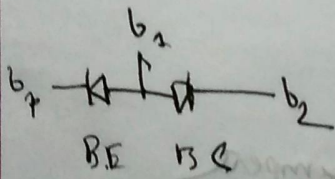
FEE-09/3/26



④

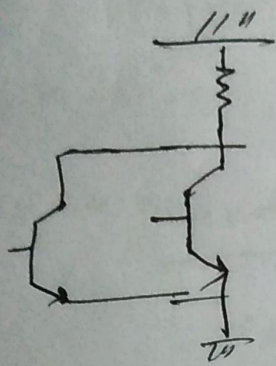


In	out
1 1 1	0
0 0 0	1



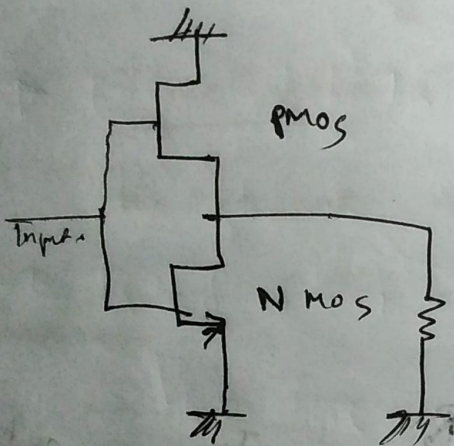
⑤ active pull up / protection

⑥ Logic diagram

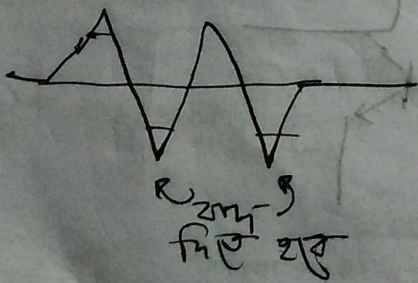


Diff pair

CMOS
PMOS
NMOS



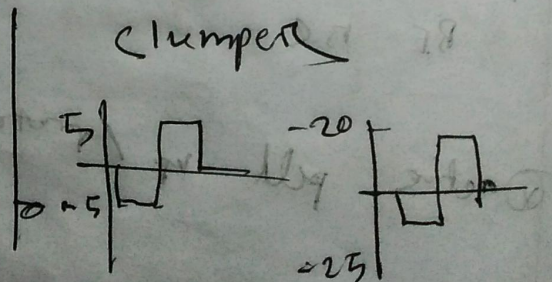
Node gate and output



analogue vs continuous

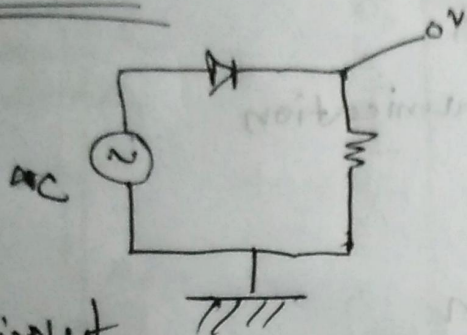
Clipper - next class

Operation amplifier



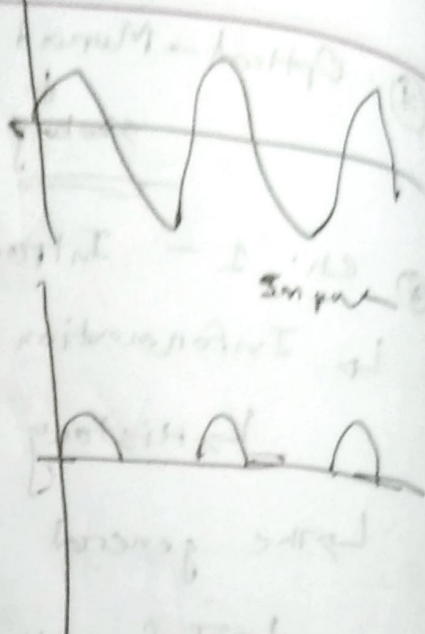
EEE - 5/17/25

Clipper:

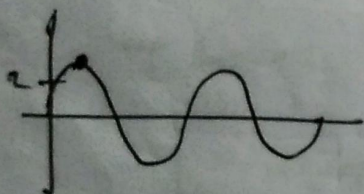


$v = V_m \sin \omega t$

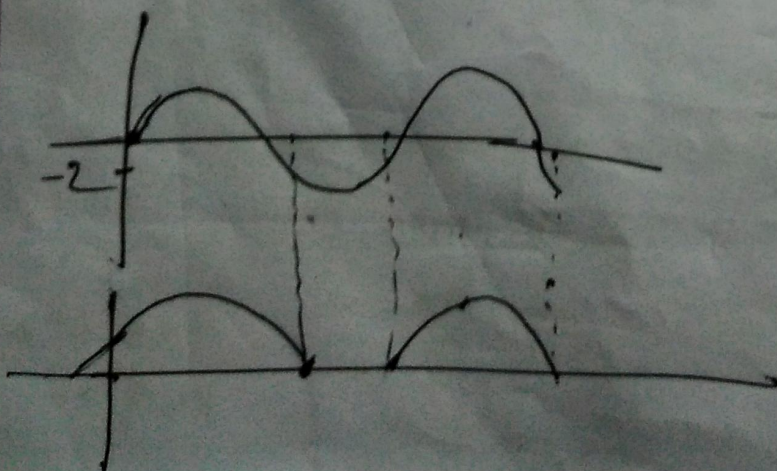
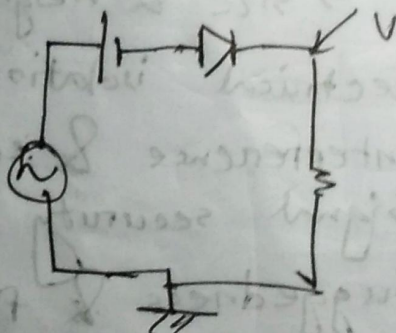
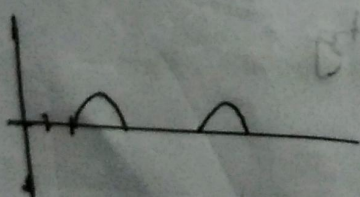
Series clipper

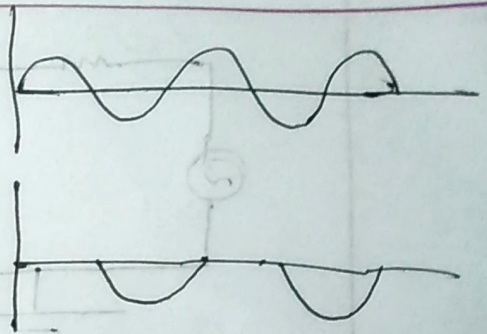
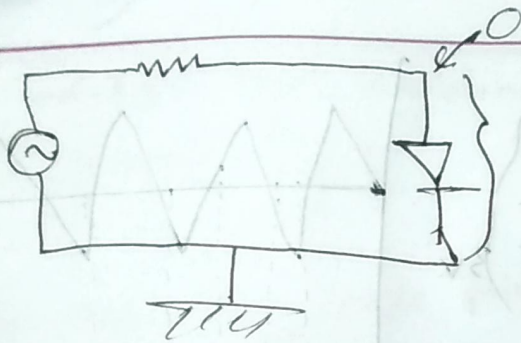


$$V = 10 \sin \omega t$$

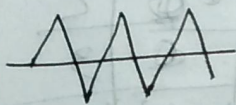
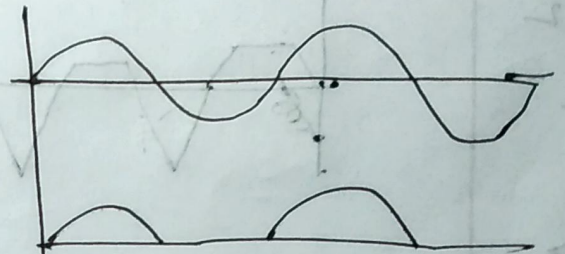
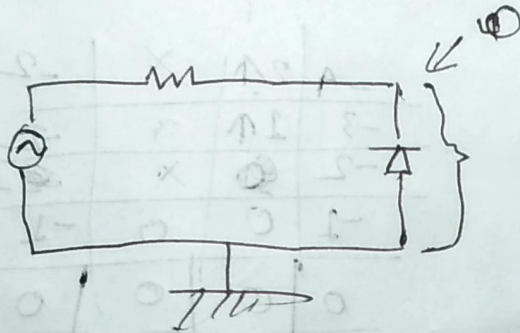


Simulation



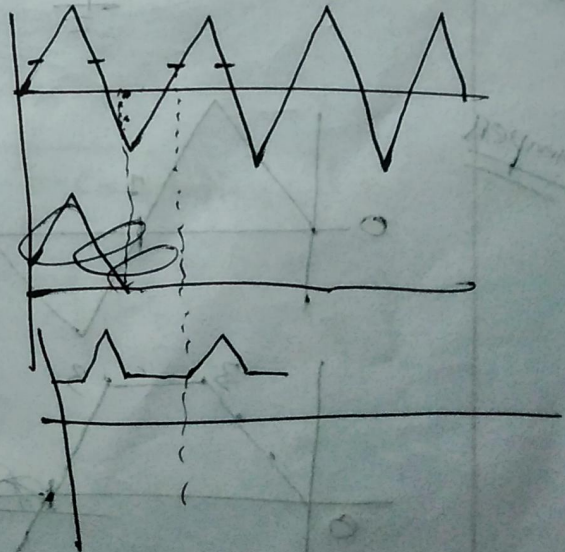
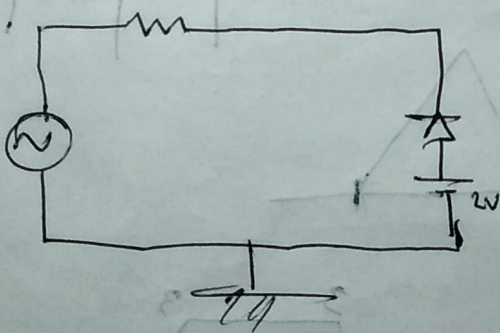


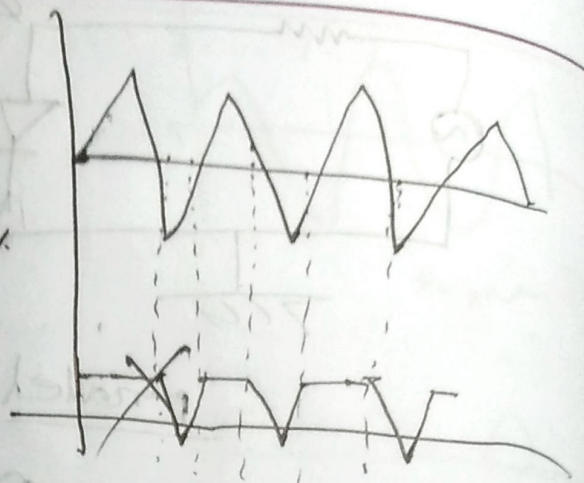
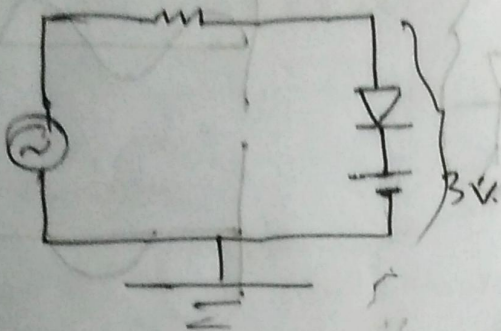
parallel clipper



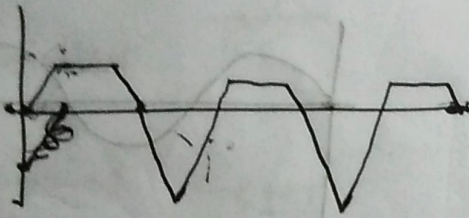
shw to all

~~sawtooth wave~~

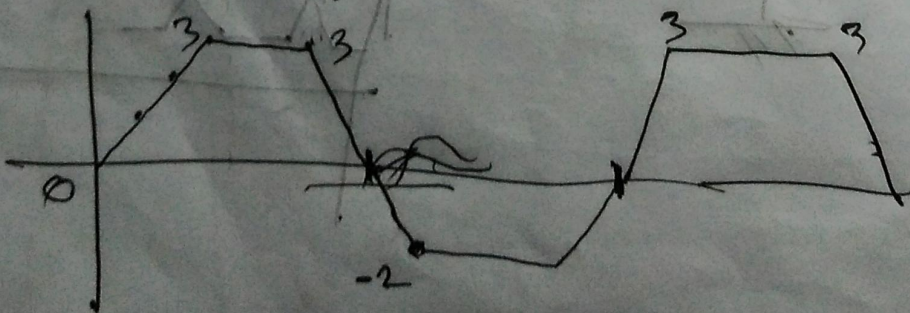
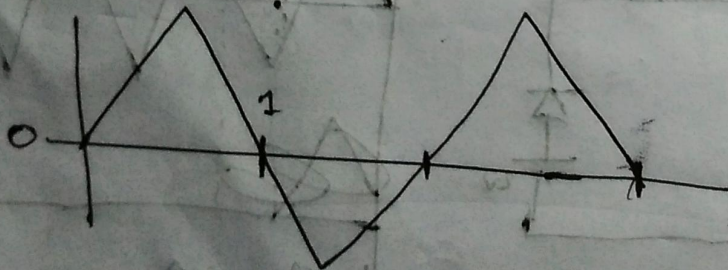
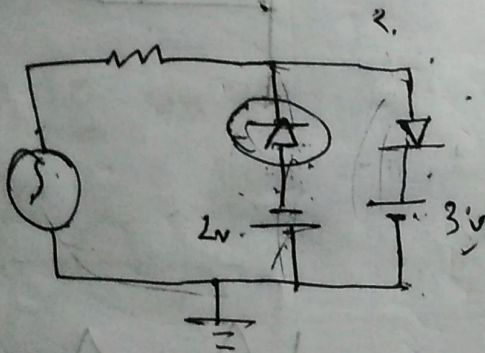




1-3
-2



-4	2↑	X	-2
-3	1↑	x	-2
-2	0	x	-2
-1	0	0	-1
0	0	0	0
1	0	0	1
2	0	0	2
3	0	0	3
4	X	1↓	3
5	x	2↓	3



clamped